Quiz 2: Probability Essentials and Random Variables

1. Which sentence best describes the role of the Partition Theorem?

(a) A partition is changing the sample space.
(b) The whole is the sum of its parts.
(c) A chain is only as strong as its weakest link.
(d) Independent events occur in the same proportions together as separately.

2. Which of the following operations corresponds to changing the sample space?

(a) intersection
(b) partitioning
(c) conditioning
(d) complement

3. Which combination of the following expressions are correct statements of the Partition Theorem?

I. $\mathbb{P}(A) = \mathbb{P}(A \mid B) + \mathbb{P}(A \mid \bar{B})$ for any events A and B.
J. $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap \bar{B})$ for any events A and B.
K. $\mathbb{P}(A) = \mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid \bar{B})\mathbb{P}(\bar{B})$ for any events A and B.
L. $\mathbb{P}(A) = \mathbb{P}(B \mid A)\mathbb{P}(B) + \mathbb{P}(\bar{B} \mid A)\mathbb{P}(\bar{B})$ for any events A and B.
M. $\mathbb{P}(A) = \mathbb{P}(A \mid B_1)\mathbb{P}(B_1) + \mathbb{P}(A \mid B_2)\mathbb{P}(B_2)$ for any events A, B_1, and B_2.

(a) I only.
(b) J, K, M only.
(c) J, K only.
(d) J, L only.

4. A medical test for diabetes is 90% accurate, but 5% of healthy patients wrongly get a positive result. Tom has received a positive result, and wants to know the probability that he has diabetes. Which one of the following formulae will be most helpful for his calculation? [Tom will need additional information once he has decided on his approach.]

(a) Bayes' Theorem
(b) Independence formula

(c) Probability of a union

(d) Partition Theorem

5. The NZ unemployment rate is 6% among those with no qualification, 4.5% among those with qualifications only at school level, and 2.1% among those with qualifications at university or other tertiary level. If I want to know the overall unemployment rate in NZ, which one of the following formulae will be most helpful? [I will need additional information to complete the calculation.]

(a) Bayes' Theorem

(b) Independence formula

(c) Probability of a union

(d) Partition Theorem

6. When a patient goes to hospital with chest pains, doctors must quickly make a decision about whether the patient is having a heart attack, based on a few visible symptoms. None of these symptoms are conclusive. Define the sample space $\Omega = \{\text{patients with chest pains}\}$, and event $H = \{\text{patient is having a heart attack}\}$. Which one of the following calculations is the most relevant?

(a) Use the partition theorem to find the probability of H by partitioning across symptoms.

(b) Find the conditional probability of H given the observed symptoms, based on previous knowledge.

(c) Use Bayes' theorem to find the probability that a heart-attack patient has these symptoms.

(d) Find the intersection probability for H and the observed symptoms.

7. In Section 2.3 of the lecture notes, we looked at an example showing 65 sons and 125 daughters in a sample of 190 children fathered by deep-sea divers. We did a hypothesis test to see whether this could happen by chance. Which one of the following statements is NOT correct?

(a) We are using the sample of 190 children to learn about the wider population of children fathered by deep-sea divers in the world.

(b) We can conceptualise a deep-sea diver repeatedly fathering children, and we want to know whether he is equally likely to produce a son or a daughter on each occasion.

(c) Our interest and conclusions must be restricted purely to the 190 children in the study.

(d) There are limits to how 'lucky' it is reasonable to be in a physical situation, determined by the laws of probability, and our aim is to test whether the data of 125/190 daughters was 'too lucky' to happen by chance.

8. A test of $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$ gives a p-value of 0.01. This means that the true value of p:

(a) is 0.5.

(b) might be 0.5 but we suspect it isn't.

(c) we don't know, but it isn't 0.5.

(d) is about 0.01.

9. If the sample data are compatible with $H_0 : p = 0.5$, it means:

(a) data like this only occur when $p = 0.5$.

(b) the data are incompatible with other values of p.

(c) data like this would be quite reasonable if p really were 0.5.

(d) data like this would be quite rare if p really were 0.5.

10. A magician asks a card player, Peter, to pick a number N between 1 and 10 inclusive, and then to deal out the first N cards from the pack. For example, if Peter picks $N = 6$, he should deal out the top 6 cards which are numbered 1, 2, 3, 4, 5, 6.
Margo does not see how many cards Peter deals out. She asks him to shuffle his N cards, and then to select just one card at random to show her. Let X be the number on the card that Peter shows Margo.
Before she sees Peter's card, Margo thinks that the probability distribution of N is $\mathbb{P}(N = n) = 1/10$ for $n = 1, 2, \ldots, 10$. Suppose she then sees Peter's card $X = 5$. Which one of the following statements is correct about her new beliefs about the probability distribution of N?

(a) Margo should still believe that all $n = 1, 2, \ldots, 10$ have equal probability.

(b) Margo should now believe that all $n = 5, 6, 7, 8, 9, 10$ have equal probability, but that n can not be 1, 2, 3, or 4.

(c) Margo should now believe that n is between 5 and 10 inclusive, but with higher probability for 5 than for 10.

(d) Margo should now believe that n is between 5 and 10 inclusive, but with higher probability for 10 than for 5.

11. Consider the following hypothesis test:

$$X \sim \text{Binomial}(n, p)$$
$$H_0 : p = 0.5$$
$$H_1 : p \neq 0.5$$

Which combination of the following statements is true?

J. For a fixed sample size n, the p-value becomes smaller as the sample proportion x/n gets further away from 0.5.

K. We will always get a smaller p-value if $x/n = 0.3$ than we will if $x/n = 0.4$, even if n differs between the two cases.

L. We will get a smaller p-value if $x/n = 0.3$ than we will if $x/n = 0.4$, if n is the same in both cases.

M. A p-value of 0.02 means that there is quite a large difference between x/n and 0.5.

(a) J, L, M only.

(b) J, L only.

(c) J, K, L only.

(d) J, K, L, M.